1 The held rough error and the closure beyond the lazy regime

Proposition 20 says where every expansion about θ_0 stops: a rough direction is fitted only after a displacement of order the analyticity radius R, so no series in $\theta - \theta_0$ describes the stretch of training in which the rough part of the error is removed. This section changes the expansion variable. The one-time theory of Part I bounds the eigenvalue of the kernel on a rough direction by the peak feature roughness, $\lambda_k R(v_k)^2 \le \sigma_R^2$ (Corollary 1), and Proposition 15 gives the exact rate of σ_R along training. Together they say that the part of the error that a roughness function counts as rough can lose energy only through one scalar function of time, $\sigma_R(t)$, up to a leak from the smooth part; and the a priori bound on the growth of σ_R then gives a floor on the rough part of the error that holds at every time, with no ball and no horizon, and that decays at most as the exponential of a cubic in t. Read across a nested family of seminorms, the floor says that the error at time t is at least the energy of the initial error above a cutoff mode that moves up as σ_R grows. An upper bound, and with it a prediction, is given in Section 2, where the expansion point is moved along the path: the one closure hypothesis an earlier draft used here (that the kernel's Rayleigh quotient on the rough residual stays a fixed fraction of σ_R^2) is measured to fail along training, and the cutoff bound of that section replaces it with a theorem and a named remainder. Everything in this section is exact.

Throughout, $R(u) = \|Pu\|$ is a projector seminorm, with P a symmetric projector on sampled functions, so that $C_R = 1$ and $\sigma_R(\theta)^2 = \lambda_{\max}\big(PK(\theta)P\big)$; the cluster seminorm and the manifold high-pass of Section 4.1 are of this form. Write

$$E_r = PE, \qquad E_s = (I - P)E, \qquad \sigma(t) = \sigma_R(\theta(t)), \tag{1}$$

for the rough and the smooth part of the error under R and for the peak feature roughness along the path. Since Y is fixed, $\dot{E}_r = P\dot{E}$, and $\|E_r(0)\| = R(E_0) \ge R(Y) - R(\hat{Y}_0)$.

1.1 The rough part loses energy only through σ_R

Proposition 1 (Energy of the rough part)**.** *Let N_θ be continuously differentiable in θ and let $\theta(t)$ be the gradient-flow path. Then, with the notation of Eq.* (1)*:*

(a) Exactly,

$$\frac{d}{dt}\,\|E_r\|^2 = -2\,\langle E_r, KE_r\rangle - 2\,\langle E_r, KE_s\rangle, \qquad 0 \le \langle E_r, KE_r\rangle \le \sigma^2\,\|E_r\|^2, \qquad |\langle E_r, KE_s\rangle| \le \sigma \tag{2}$$

(b) For every $\varepsilon > 0$ and every t,

$$\|E_r(t)\|^2 \;\ge\; \|E_r(0)\|^2\,\exp\Big(-(2+\varepsilon)\int_0^t \sigma(s)^2\,ds\Big) - \frac{1}{\varepsilon}\int_0^t \lambda_1(s)\,\|E_s(s)\|^2\,\exp\Big(-(2+\varepsilon)\int_s^t \sigma(\tau)^2 \tag{3}$$

(c) The rate at which the rough part can lose energy is bounded by the peak feature roughness alone once the smooth part is gone: $-\frac{d}{dt}\|E_r\|^2 \le 2\sigma^2\|E_r\|^2 + 2\sigma\lambda_1^{1/2}\|E_r\|\,\|E_s\|$.

Proof. (a) $\frac{d}{dt}\frac{1}{2}\|E_r\|^2 = \langle PE, P\dot{E}\rangle = \langle E_r, \dot{E}\rangle = -\langle E_r, KE\rangle$ by Theorem 7(a), and $E = E_r + E_s$. K is positive semidefinite, so $\langle E_r, KE_r\rangle \ge 0$. For a unit ψ in the range of P, $R(\psi) = \|\psi\| = 1$, so $\psi \in R^\circ$ and $\|\psi\|_K \le \sigma_R$ by the definition of σ_R (proof of Theorem 1(a)); hence $\langle E_r, KE_r\rangle = \|E_r\|_K^2 \le \sigma^2\|E_r\|^2$. For the cross term, $\langle a, Kb\rangle = g_a \cdot g_b$, so $|\langle E_r, KE_s\rangle| \le \|E_r\|_K\|E_s\|_K \le \sigma\|E_r\| \cdot \lambda_1^{1/2}\|E_s\|$. (b) By (a) and $2ab \le \varepsilon a^2 + b^2/\varepsilon$ with $a = \sigma\|E_r\|$ and $b = \lambda_1^{1/2}\|E_s\|$,

$$\frac{d}{dt}\|E_r\|^2 \;\ge\; -(2+\varepsilon)\,\sigma^2\,\|E_r\|^2 \;-\; \frac{\lambda_1}{\varepsilon}\,\|E_s\|^2,$$

a linear differential inequality for the nonnegative function $\|E_r\|^2$; Grönwall's inequality in its integrating-factor form gives Eq. (3). (c) is (a) with the two bounds. □

Part (a) is the energy identity of Theorem 7(a) projected on the rough subspace. Its content is that the kernel enters the energy of the rough part only through two numbers at each instant: its Rayleigh quotient on the rough subspace, capped by σ^2, and its coupling of the rough subspace to the smooth one, capped by $\sigma\lambda_1^{1/2}$. Part (b) is the dynamic form of the floor of Theorem 4. Where Theorem 8(b) charges the path linearly, $\int \sigma_R|\dot{\theta}|$, Eq. (3) charges it exponentially, $\exp(-(2+\varepsilon)\int\sigma^2)$, and the two are different readings of the same fact, that the output can acquire roughness only at rate σ_R. The second term is the leak: the smooth part of the error drives the rough part through the off-diagonal block of the kernel, and it is small once the smooth part has been removed, which the floor of Part I says happens

while the rough part stays. With $\varepsilon \to 0$ the exponent tends to the rate $2\sigma^2$ of Corollary 1 and the leak coefficient grows; for a bound that is to be read along a run, ε is chosen to balance the two terms. Along a run, $\sigma(t)$ is a top eigenvalue of PKP at each saved state and $\|E_s(t)\|$ is read off, so Eq. (3) is a check with no free constant.

1.2 The a priori floor on the rough part

Proposition 15(b) bounds the growth of σ_R along gradient flow by $\Theta_R \le \sigma_R \lambda_1^{1/2} H_2 \|E\|$ (with $C_R = 1$). Inserted into Proposition 1, this gives a floor on the rough part that is computed from the initial state and holds at every time.

Theorem 1 (The rough part is held: a priori form)**.** *Let N_θ be twice continuously differentiable in θ, let $\theta(t)$ be the gradient-flow path, and let $\bar\lambda_1$ and $\bar H_2$ be the largest values of $\lambda_1(\theta(s))$ and $H_2(\theta(s))$ for $s \le t$ (or over the ball of radius $\ell(t)$ about θ_0, which contains the path). Let $\sigma_0 = \sigma_R(\theta_0)$ and*

$$a = \bar\lambda_1^{1/2}\, \bar H_2\, \|E_0\|, \qquad \Sigma(s) = \sigma_0 + a\, s, \qquad I(t) = \int_0^t \Sigma(s)^2\, ds = \sigma_0^2\, t + \sigma_0\, a\, t^2 + \tfrac{1}{3}\, a^2\, t^3. \tag{4}$$

Then $\sigma(s) \le \Sigma(s)$ for all $s \le t$, and for every $\varepsilon > 0$,

$$\|E_r(t)\|^2 \;\ge\; \|E_r(0)\|^2\, e^{-(2+\varepsilon) I(t)} \;-\; \frac{\bar\lambda_1}{\varepsilon}\, J_s(t), \qquad J_s(t) = \int_0^t \|E_s(s)\|^2\, ds. \tag{5}$$

In particular $\|E(t)\| \ge \|E_r(t)\|$ obeys the same bound, and $\|E_r(0)\| \ge R(Y) - R(\hat Y_0)$.

Proof. Let $S(t) = \sigma_0^2 + 2\int_0^t \Theta_R$, so that $\sigma(t)^2 \le S(t)$ by Eq. (98) and $S' = 2\Theta_R \le 2\sigma\lambda_1^{1/2} H_2 \|E\| \le 2 S^{1/2} \bar\lambda_1^{1/2} \bar H_2 \|E_0\|$, using $\sigma \le S^{1/2}$ and $\|E\| \le \|E_0\|$ (Theorem 7(a)). Hence $(S^{1/2})' \le a$ and $\sigma(s) \le S(s)^{1/2} \le \sigma_0 + as = \Sigma(s)$. Insert $\sigma \le \Sigma$ into Eq. (3), bound $\lambda_1(s)$ by $\bar\lambda_1$ and the inner exponential in the leak term by one. □

The theorem is the counterpart, for the energy of the rough part, of the minimum-path-length statement of Corollary 5. Three features distinguish it. It holds at every time with no horizon and no analyticity, since nothing is expanded. Its decay is at most exponential in a cubic: with no rough features at the start, $\sigma_0 = 0$, which is the situation of Proposition 20 for a network with a smooth activation, the exponent is $\frac{2+\varepsilon}{3} a^2 t^3$, so the rough

part is held for a time of order $a^{-2/3}$. And the kernel enters only through σ_0, $\bar{\lambda}_1$ and $\bar{H}_2$: the initial peak feature roughness, the top eigenvalue, and the second-derivative size, which are the three numbers through which an architecture enters the rest of this section. The quantity $J_s(t)$ is the time integral of the energy of the smooth part; a priori it is bounded by the band of Theorem 10 about $e^{-K_0 t}E_0$ projected on $I-P$, and along a run it is read off.

Corollary 1 (Minimum time to remove the rough part)**.** *Under the hypotheses of Theorem 1, if* $\|E_r(t)\| \le \epsilon$ *with* $\epsilon^2 + \bar{\lambda}_1 J_s(t)/\varepsilon < \|E_r(0)\|^2$*, then*

$$(2+\varepsilon)\, I(t) \;\ge\; \log \frac{\|E_r(0)\|^2}{\epsilon^2 + \bar{\lambda}_1 J_s(t)/\varepsilon}, \tag{6}$$

and in particular, when $\sigma_0 = 0$*,*

$$t \;\ge\; \Big(\frac{3}{(2+\varepsilon)\, a^2}\, \log \frac{\|E_r(0)\|^2}{\epsilon^2 + \bar{\lambda}_1 J_s(t)/\varepsilon}\Big)^{1/3}. \tag{7}$$

Proof. Rearrange Eq. (5); I is increasing in t, and for $\sigma_0 = 0$, $I(t) = a^2 t^3/3$. □

Eq. (7) is the a priori ignition time of a rough direction from a smooth start, and it depends on the architecture through $a = \bar{\lambda}_1^{1/2}\bar{H}_2\|E_0\|$ alone: the fastest the peak feature roughness can grow per unit time is $\bar{\lambda}_1^{1/2}\bar{H}_2$ per unit of error, and the rough part is held until the integrated square of that growth is of order one. Like Corollary 5 it is one-sided; unlike it, the time scale is a cube root rather than a square, because the roughness supply itself grows linearly while the error is held.

1.3 The roughness spectrum and the moving cutoff

The floor of Theorem 1 holds for every projector seminorm. Read across a nested family, it locates where the held error sits. Let $P_1 \succeq P_2 \succeq \cdots$ be a nested family of projectors, each the projector of a seminorm $R_m(u) = \|P_m u\|$; for the manifold high-pass of Section 4.1, P_m removes the first m graph-Laplacian modes, and the range of P_m shrinks as m grows. Write $\sigma_m(\theta)^2 = \lambda_{\max}(P_m K P_m)$, which is nonincreasing in m, and

$$\rho_m = \|P_m E_0\|, \qquad I_m(t) = \sigma_m(\theta_0)^2\, t + \sigma_m(\theta_0)\, a\, t^2 + \tfrac{1}{3}a^2 t^3. \tag{8}$$

$m \mapsto \rho_m$ is the roughness spectrum of the initial error: the energy of E_0 above mode m. It is fixed by the dataset and the initial output, and for a target with $\hat{Y}_0$ small it is the roughness spectrum of the labels.

Definition 1 (Cutoff mode)**.** *For $t \geq 0$ and $\varepsilon > 0$, the cutoff mode is*

$$m^\star(t) = \min\{\, m : (2+\varepsilon)\, I_m(t) \leq \log 2 \,\}, \tag{9}$$

the lowest mode above which, at time t, the a priori bound allows the rough part to have lost at most half its energy.

Since I_m is nonincreasing in m and increasing in t, the set in Eq. (9) is an up-set in m and $m^\star(t)$ is nondecreasing in t: the cutoff moves up through the modes as training proceeds, and it moves even when $\sigma_m(\theta_0) = 0$, through the a^2t^3 term, which is feature learning.

Proposition 2 (The error at time t is at least the energy above the cutoff)**.** *Under the hypotheses of Theorem 1, for every t,*

$$\|E(t)\|^2 \;\geq\; \|P_{m^\star(t)}E(t)\|^2 \;\geq\; \tfrac{1}{2}\,\rho^2_{m^\star(t)} \;-\; \frac{\bar\lambda_1}{\varepsilon}\int_0^t \|(I - P_{m^\star(t)})E(s)\|^2\, ds. \tag{10}$$

Proof. Theorem 1 with $P = P_{m^\star(t)}$ and $e^{-(2+\varepsilon)I_{m^\star}(t)} \geq \frac{1}{2}$ by the definition of $m^\star$. □

In words: at time t, the error is at least the part of the initial error that is rougher than the cutoff mode, up to the leak, and the cutoff is set by the roughness capacity of the network, $\sigma_m(\theta_0)$ and a, and by nothing else. Without feature learning ($a = 0$) the cutoff is the smallest m with $\sigma_m(\theta_0)^2 t \leq \log 2/(2+\varepsilon)$, which is the spectral bias of the lazy regime stated in roughness rather than in frequency; with feature learning the cutoff rises faster, and Corollary 1 says how much faster it can rise at most. This is the quantity that the closure below takes as its estimate of the held rough error: for a dataset whose roughness spectrum has a step at some mode m_Y, the error is held near ρ_{m_Y} until $m^\star(t)$ passes m_Y, and the time at which it does is Eq. (6) for $P = P_{m_Y}$.

1.4 The closure, revisited

The statements above bound the rough part from below. The upper bound needs the growth of the kernel *along the residual* with the sign that learning needs, and the natural closure, that $\langle E_r, KE_r\rangle \geq c\,\sigma_R^2\|E_r\|^2$ with c of order one, is false along training: measured on four runs with the manifold high-pass at $m = 20$ and 60, the ratio $\langle E_r, KE_r\rangle/(\sigma_R^2\|E_r\|^2)$ is 0.02–0.09 at initialisation and decays to 0.001–0.005, while σ_R^2 is lifted 6–50× [n=200].

The mechanism is the one the chain of Section 2 proves: the kernel's growth is targeted at the smoothed residual and raises the top of the rough subspace by the gain law, while the residual itself moves to the slowest directions of that subspace, so $c(t) \approx \lambda_{\text{eff}}(t)/\sigma_R(t)^2 \to 0$. The error is not held in any fixed roughness seminorm; what is held is its energy on the modes still unfit in the *current* kernel basis. The two-sided statement that replaces the band is the cutoff bound of Section 2: the energy of the initial error on the continuously transported unfit set, with the cross-front rotation as the only remainder, and the onset of the lift given by a convergent expansion in the richness parameter in place of a reduced model.

1.5 Depth and the comparison of architectures

Everything above enters an architecture through three numbers at the start of training, $\sigma_R(\theta_0)$, $\bar{\lambda}_1$ and $\bar{H}_2$, and through the two run quantities c and Θ_R. For two designs on the same dataset, the comparison principle of Corollary 6 applies with the bounds of this section.

Corollary 2 (Comparing designs on the rough part)**.** *Let designs A and B be trained by gradient flow on the same dataset, let P be the projector of a roughness function fixed by the inputs, and let $T > 0$.*

(i) (Proved.) $\|E_r^A(T)\|^2$ is at least the right side of Eq. (5) *computed with σ_0^A, a_A and $J_s^A(T)$.*

(ii) (Proved, Section 2.) $\|E^B(T)\|^2$ is at most the ceiling of the cutoff bound computed for B: the energy of its initial error on the modes that are still unfit at T under its lifted spectrum, plus the cross-front leakage and the decayed fit part.

(iii) If the bound in (ii) is below the bound in (i), then $\|E^B(T)\| < \|E_r^A(T)\|$: B has learned more of the rough part of the labels by time T.

The corollary separates what depth does into two effects that can be checked one at a time. A deeper network can have larger initial peak feature roughness $\sigma_R(\theta_0)$, because its gradient features decorrelate over shorter input distances when the initialization is near the edge of chaos; that lowers the held rough error after the lazy phase, since the cutoff of Definition 1 starts higher. And it can have larger $\lambda_1^{1/2} H_2$, which is the rate b at which it can supply roughness per unit of error; that shortens the ignition time of Corollary 1 and advances the onset of the lift given by the richness expansion of Section 2. Both are computed before training, one top eigenvalue

of PKP and one nested automatic differentiation per design, and both are expected to grow with depth; the theory says which part of the error each one removes and in what time, but not that either grows with depth, which is a measurement or a statement about the initialization. Where the initialization statistics of deep networks are known in closed form, they can supply $\sigma_R(\theta_0; L)$, $\lambda_1(L)$ and $H_2(L)$ as functions of depth, and this is the only place in the theory where a limit of the initialization enters; nothing along the path is taken to a limit.

1.6 Scope and what to measure

Exact under gradient flow, no hypothesis beyond smoothness. Proposition 1; Theorem 1 and Corollary 1 with $\bar{\lambda}_1$ and $\bar{H}_2$ taken along the path or over the ball of radius $\ell(t)$; Proposition 2; Corollary 2(i). The leak term is exact and is read off a run; its a priori bound through Theorem 10 is available while the band of that theorem is informative. **Proved in Section 2.** The cutoff bound (two-sided, with the leakage as remainder); Corollary 2(ii)–(iii); the onset law. Nothing in this section is approximated; the one hypothesis of the earlier draft has been measured false and removed. The measurements that decide whether the closure is the right one, to be added to the list of Section 15: (13) $\sigma_R(t)$ along each run for the manifold high-pass at several m, against its a priori bound $\Sigma(t)$ of Eq. (4) and against the realized drive Θ_R; (14) the ratio $\langle E_r, KE_r\rangle/(\sigma_R^2\|E_r\|^2)$ along each run, which was the constant c of the earlier closure and is now a diagnostic of where the residual sits in the rough subspace (measured: decaying to 10^{-3} [n=200]); (15) $\|E_r(t)\|^2$ against Eq. (3) and against the two sides of the cutoff bound of Section 2, and $J_s(t)$; (16) the cutoff mode $m^\star(t)$ against the mode at which the measured error's roughness spectrum actually drops, which tests Proposition 2 as an estimate rather than as a bound; (17) $\sigma_R(\theta_0)$, λ_1 and H_2 as functions of depth at initialization, which are the three numbers through which depth enters Corollary 2.

2 Training through time: the chained expansion and the limits of prediction

Proposition 20 says that no series in $\theta - \theta_0$ reaches the stretch of training in which the rough part of the error is removed, and Section 1 replaced the expansion variable by the peak feature roughness. This section keeps θ as the variable and moves the expansion point instead: training is written as

a chain of expansions, each about the current state and each valid within its own radius, whose composition is exact. The chain gives three kinds of statement. Identities that hold at every time for any network in a broad class (§2.1); structure theorems for what gradient flow does to the weights and the kernel at each stage, which is where the low-rank "spikes" and the targeted growth of the kernel come from (§2.3–2.4); and applications in which an open question becomes a computation with the chain or a theorem about its leading orders (§2.5). The last two subsections state why the late stage cannot be predicted from the initialisation without carrying the state forward, in a form that is a theorem rather than an impression (§2.7), and what a statistical description can and cannot promise (§2.8).

2.1 Setting and exact identities

Model class. N_θ is any composition of *parameter-linear blocks* $z_\ell = W_\ell a_\ell$, where a_ℓ is whatever the block receives (dense, convolution, attention projections, normalisation gains), and *parameter-free maps* between them (pointwise activations, softmax, normalisation), differentiable almost everywhere in θ. ReLU qualifies. Training is gradient flow on the squared loss, or any update of the data-indexed outer-product form below (GD, SGD on a subset of the data, momentum through its integral); Adam is treated in §2.5. Write $\Psi(t) = J(\theta(t)) \in \mathbb{R}^{N\times P}$ for the Jacobian on the n data points ($N = n\,n_y$), $K = \Psi\Psi^\top/n$, $\delta^c_\ell(x) = \partial \hat{Y}_c(x)/\partial z_\ell$ for the backprop signal into block ℓ, and $U(t) = \int_0^t E$ for the integrated residual.

Proposition 3 (Four identities)**.** *For every t at which N_θ is differentiable:*

(a) $\dot{E} = -K(t)E$*; hence* $\frac{d}{dt}\|E\|^2 = -2\langle E, KE\rangle$ *and* $E(T) = \mathrm{Texp}\big(-\int_0^T K\big)E(0)$.

(b) $K = \frac{1}{n}\sum_\ell A_\ell \odot D_\ell$*, with* $A_\ell[i,j] = a_\ell(x_i)\cdot a_\ell(x_j) + 1$ *the Gram of the block's inputs and* $D_\ell[(i,c),(j,d)] = \sum_q \delta^c_{\ell,q}(x_i)\,\delta^d_{\ell,q}(x_j)$ *the Gram of its backprop signals.*

(c) $W_\ell(t) - W_\ell(0) = -\frac{1}{n}\int_0^t \sum_i \sum_c \delta^c_\ell(x_i)\,E_c(x_i)\,a_\ell(x_i)^\top$*: the change of every block is a sum of outer products indexed by the data, its row space lies in the span of the block's inputs on the data, and every block is driven by the same residual.*

(d) For a unit q of block ℓ with preactivation z_q and incoming backward signal $m_q = W_{\ell+1}^\top \delta_{\ell+1}$,

$$\dot{z}_q = \dot{W}_\ell a_\ell + W_\ell\,\dot{a}_\ell, \qquad \dot{m}_q = \dot{W}_{\ell+1}^\top \delta_{\ell+1} + W_{\ell+1}^\top \dot{\delta}_{\ell+1}.$$

The first terms are given by (c) (the coherent *part); the second carry the motion of the neighbouring blocks through the current weights (the* propagation *part). Together with (a) and (b) this is a closed system on the data.*

Proof. (a) is the chain rule for $\theta' = -\frac{1}{n} J^\top E$. (b): $\partial \hat{Y}_c(x)/\partial W_\ell[q,p] = \delta^c_{\ell,q}(x)\, a_{\ell,p}(x)$. (c): integrate the gradient. (d): differentiate the definitions. □

Remark 1. Everything in this section is a consequence of Proposition 3. In particular the propagation part of (d) is the only place where the initialisation enters other than through the unit statistics; omitting it (as any model built from the kernel alone must) was measured to cost 11–28% of the unit displacements early and a factor 2 in the late error on the reference problem [n=200].

2.2 The stage identity and the chain

Fix stage boundaries $0 = t_0 < t_1 < \cdots < t_S = T$ and write $\delta\Psi_s = \Psi(t_s) - \Psi(t_{s-1})$.

Proposition 4 (Stage identity)**.** *$K(t_s) - K(t_{s-1}) = \mathrm{Cross}_s + Q_s$ with $Q_s = \delta\Psi_s \delta\Psi_s^\top/n \succeq 0$ and $\mathrm{Cross}_s = (\Psi\delta\Psi_s^\top + \delta\Psi_s\Psi^\top)/n$, $\Psi = \Psi(t_{s-1})$. For a unit vector v, the quadratic form changes by $\frac{1}{n}\big[2\langle \Psi^\top v, \delta\Psi_s^\top v\rangle + \|\delta\Psi_s^\top v\|^2\big]$, which is positive whenever $\|\delta\Psi_s^\top v\| > 2\|\Psi^\top v\| = 2\sqrt{n\lambda_v}$.*

Theorem 2 (Feature-sum identity and bracket at T)**.** *$\Psi_T = \Psi_0 + \sum_s \delta\Psi_s$, so for every unit v, with $D_v = \sum_s \delta\Psi_s^\top v$ and $\ell_v = \|\Psi_0^\top v\|$,*

$$\tfrac{1}{n}\big(\|D_v\| - \ell_v\big)_+^2 \;\le\; v^\top K_T v \;\le\; \tfrac{1}{n}\big(\|D_v\| + \ell_v\big)^2.$$

D_v splits exactly into coherent and propagation parts by Proposition 3(d). On the directions the residual has driven, ℓ_v is negligible and the lower bound is the value: measured 0.90–1.00 *of $v^\top K_T v$ for every rank ≥ 60 on the reference problem [n=200]. Where J is C^1 along the path, $|\lambda_k(K_T) - \lambda_k(K_0)| \le \int_0^T C(s)\|E(s)\|\, ds$ with $C = \frac{2}{n^2}\|\Psi\|_2^2\|\Psi'\|_2$ (Weyl), so every lifted eigenvalue at T is bracketed between two functionals of the residual history.*

Theorem 3 (Stage error and the chain)**.** *Where J is C^2 along the path, $\delta\Psi_s = \alpha_s\dot{\Psi}(t_{s-1}) + R_s$ with $\|R_s\| \le \frac{\alpha_s^2}{2}\sup\|\ddot{\Psi}\|$; choosing $\alpha_s\|\ddot{\Psi}\| \le \varepsilon\|\dot{\Psi}\|$ makes each stage's first-order increment accurate to relative ε, and the composition of the stages is the exact trajectory. The quantity $\|\ddot{\Psi}\|/\|\dot{\Psi}\|$ is the inverse of the stage length over which the linearisation at the current state is trustworthy; its norm along the descent direction is H_2.*

2.3 What a stage does to the weights: the spike theorem

For a hidden block whose outgoing weights are $W_{\ell+1}$, write $S[i,q] = \sigma'(z_q(x_i))$ for the local derivative of the map following the block, $M[i,q]$ for the backprop coefficient contracted with the drive (for the last hidden block $M = U\,W_{\text{out}}$, of rank $\le n_y$), and A for the block's inputs on the data. By Proposition 3(c), $\Delta W = -\frac{1}{n}(S \odot M)^\top A$.

Theorem 4 (Core and bulk)**.** *(a) (*at initialisation*) Split $S = h_0 + (S - h_0)$ with h_0 the mean of σ' over the data. Then $\Delta W = \Delta W^{\text{core}} + \Delta W^{\text{bulk}}$ with $\Delta W^{\text{core}} = -\frac{h_0}{n} M^\top A$ of rank $\le n_y$, its left singular space spanned by the rows of W_{out} and its right singular space by $A^\top U$. If the block's input weights are independent of $W_{\ell+1}$ (true at θ_0), the expected bulk-to-core energy ratio is $\rho = \mathrm{Var}(\sigma'(z))/\mathbb{E}[\sigma'(z)]^2$ at the preactivation scale. The output block has rank $\le n_y$ and the input block rank $\le d_{\text{in}} + 1$ exactly.*

*(b) (*any time, no independence*) For any rank-r approximation S_r of S with remainder $\tilde{S}$, $\mathrm{rank}\big((S_r \odot M)^\top A\big) \le n_y r$ and $\|(\tilde{S} \odot M)^\top A\|_F \le \|\tilde{S}\|_{\max}\|M\|_F\|A\|_2/n$. Since $Z = A\,W_\ell^\top + b$, the numerical rank of $S = \sigma'(Z)$ is controlled by the spectrum of the block's input Gram and by the Hermite coefficients of σ' at the current preactivation scale, both quantities the chain carries; for deeper blocks the bound composes multiplicatively down the stack.*

Proof. (a) Expand σ'; the core is $h_0 M$; the bulk's expected energy factorises by independence of S_{iq} and M_{iq}. (b) $\mathrm{rank}(X \odot Y) \le \mathrm{rank}X \cdot \mathrm{rank}Y$, the entrywise Frobenius bound, the column-space inclusion for Z, and the Hermite expansion. □

2.4 The richness expansion, its radius, and the chain as its resummation

Let ϵ be any parameter that multiplies the weight velocity at fixed residual dynamics in a suitable time: the inverse output scale $1/s$ with $\tau = s^2 t$ for centred outputs $\hat{Y} = s(N_\theta - N_{\theta_0})$; the Euler step Δ of §2.5; a μP-type multiplier; an inverse-width factor. Then

$$\frac{dE}{d\tau} = -K(\theta)E, \qquad \frac{d\theta}{d\tau} = -\frac{\epsilon}{n} J(\theta)^\top E.$$

Theorem 5 (All-orders richness expansion)**.** *For analytic parameter-free maps and each fixed τ there is $R(\tau) > 0$ such that every analytic observable of the state is a convergent power series in ϵ for $|\epsilon| < R(\tau)$; in particular*

$$\frac{v^\top K(\tau) v}{v^\top K(0) v} - 1 = \sum_{k \geq 1} a_k(\tau)\, \epsilon^k,$$

with coefficients determined recursively from θ_0 and the lazy trajectory alone: a_0 is kernel regression with $K(\theta_0)$, $\theta^{(1)}(\tau) = -\frac{1}{n} J_0^\top U_{\mathrm{lazy}}(\tau)$, $a_1 = 2\langle \Psi_0^\top v, J'[\theta^{(1)}]^\top v \rangle / (n\lambda_v)$, and a_k involves derivatives of J at θ_0 up to order k along lower-order displacements and the lower-order residual corrections (linear inhomogeneous equations, variation of constants in the eigenbasis of K_0). No quantity along a trained path enters at any order.

Theorem 6 (Radius, and the chain as resummation)**.** *$R(\tau)$ is bounded below by the stage criterion of Theorem 3 applied at θ_0, and along a direction v the series is valid while $\|\delta\Psi^\top v\| < \|\Psi_0^\top v\|$ (Proposition 4). Where this fails the lift along v is of order one and no finite or infinite number of terms of the series about θ_0 represents it. Re-expanding at each stage boundary (Theorem 3) continues the series past $R(\tau)$ by moving the expansion point along the path; within $R(\tau)$ the two agree, beyond it only the chain is defined.*

Proof. Analytic dependence of ODE solutions on parameters; the recursion is the Taylor expansion of the vector field in ϵ; the radius statement is Proposition 20 at the expansion point θ_0 together with Proposition 4. □

2.5 Applications

Euler versus direct. Let $y_D = \Phi_\Delta(x)$ and $y_E = (\Phi_\Delta(x) - x)/\Delta$ be the direct and the Euler target, trained from the same θ_0 with centred outputs, and write the direct residual on the increment as $E_D = \Delta \tilde{E}$.

Theorem 7. *$\tilde{E}$ and E_E obey the same residual equation with their own kernels, start from the same value, and the weight velocities satisfy $\dot{\theta}_D = -\frac{\Delta}{n} J^\top \tilde{E}$ against $\dot{\theta}_E = -\frac{1}{n} J^\top E_E$: the direct problem is the Euler problem with the richness parameter multiplied by Δ (Theorem 5 evaluated at a smaller argument). Hence (i) direct's stage lifts on the increment are Δ^2 times Euler's at equal normalised residual; (ii) as $\Delta \to 0$ at fixed budget the direct network is in the lazy regime on the increment, its normalised error converges to the frozen-kernel error of the increment target, and the held-rough bound of Section 1 becomes tight for it; (iii) Euler's error is the full feature-learning error, below that bound by the feature-learning gain, which grows*

with the budget and with the increment's roughness. The direct network additionally spends a Δ-independent early drive on the identity component of its target.

The static argument of the previous sections ("the increment is a smoother target") is the frozen-kernel shadow of this: dividing by Δ multiplies the drive by $1/\Delta$ and the lifts by $1/\Delta^2$, so Euler trains rich on the increment while direct trains lazy on it. Statement (ii) is exact; (iii) is conditional on the rate comparison below.

Progressive sharpening, the initial reduction phase, and the edge of stability. By Proposition 4, $\dot{\lambda}_1 = v_1^\top(\text{Cross}+Q)v_1$ per stage with $Q \succeq 0$. Along the residual the cross term at $t = 0$ equals $-\frac{2\alpha}{n^2}\langle E, N''_\theta[g,g]\rangle$ with $g = J^\top E$, exactly, and it is negative on 90–91% of the eigen-directions for the raw residual $E = \hat{Y}_0 - Y$ and on 48–55% (no systematic sign) for the centred residual $-Y$ [n=200]: the initial sharpness-*reduction* phase reported in the warmup literature is the network unlearning its own initial function and is absent for centred initialisations. Once learning has accumulated, the coherent part of the cross term is the inner product of the accumulated core above with the stage core below through the shared drive, positive while the drive keeps its direction (measured positive on 80–100% of modes from $t \approx 1$ on): progressive sharpening, faster with depth by composition ($5\times, 11\times, 20\times, 86\times$ top gain at depths $2, 3, 6, 6$ [n=200]). Taking GD steps as stages, above the stability threshold the top component of the residual alternates in sign, the stage cores along it alternate, their cross terms with the accumulated core are negative at the core level, and λ_1 decreases until the step is stable again: sharpening and self-stabilisation are the two signs of one term. What is not proved is the sign of the init-feature part of the cross term after the transient (Lemma 1 below).

Adam. For $\dot{\theta} = -P(t)\nabla L$ with P diagonal positive, (a) of Proposition 3 holds with $K_P = \Psi P \Psi^\top/n$, (c) with the outer products multiplied entry-wise by P_ℓ, and Proposition 4 and Theorem 2 verbatim with $P^{1/2}$-weighted features. The spike statement needs one more fact: P is built from the gradient block itself, so $P \odot G$ is a normalised sign pattern of a rank-$\leq n_y$ matrix, and the same mean-plus-fluctuation argument as Theorem 4 gives its core fraction as the squared first Hermite coefficient of the normaliser under the entry distribution of G: $2/\pi$ for Gaussian-like entries, 1 for coordinate-aligned (sparse) singular vectors, with the time-averaged second moment

lowering it further (measured 0.68–0.81, 0.95–1.00, 0.52 respectively on synthetic blocks). This is the rotation-dependent property that distinguishes Adam from SGD.

Validation. With J_{val} the Jacobian on validation points, $\dot{\hat{Y}}_{\text{val}} = -K_{vt}E$ with $K_{vt} = J_{\text{val}}J^\top/n$, exactly, so $-\frac{d}{dt}\frac{1}{2}\|E_{\text{val}}\|^2 = E_{\text{val}}^\top K_{vt}(0)E + E_{\text{val}}^\top[K_{vt}(t) - K_{vt}(0)]E$: a lazy-transfer part and a feature-learning-transfer part. The coherent part of the lift on validation points is the training stage core transported by the cross activation Gram; under equal unit statistics on the two samples the whole chain applies with that substitution. Measured [n=200]: early both parts positive and the validation rate 70–110% of the training rate; late the lazy part negative (the frozen kernel maps a residual living in sample-specific rough modes into directions that increase validation error, the static held-rough mechanism seen as a sign) while the lifted part stays positive. At the sample sizes of interest, where training and validation agree, the sign pattern is a prediction to be checked rather than a result.

Delayed feature learning ("grokking"). The mechanism is Theorems 4–2: the residual drives a positive lift along the directions it has visited, so a direction too slow for the initial kernel to fit within the budget is fit later through the lift. The onset is Theorem 5 with $\epsilon = 1/s$: an init-computable series, verified to 0–16% in its range, with a proved radius; the delay to a small lift is its inversion, and because a_1 changes sign no universal exponent in s exists, which accounts for the disagreement among reported scalings. The event itself, a lift of order one or more along a misaligned direction, lies outside $R(\tau)$ by construction and is a chain computation; by §2.7 it is not predictable from the initialisation in any other way. Rigorous timings in the literature (weight decay, modular arithmetic, two-layer mean field) are special cases evaluated inside their own radii.

The error at T, term by term. Write $K(t) = K_0 + \Delta(t)$ with Δ the accumulated stage increments. The residual equation has the exact Dyson form

$$E(T) = \sum_{m\geq 0}(-1)^m \int_{T>u_1>\cdots>u_m>0} e^{-K_0(T-u_1)}\Delta(u_1)\, e^{-K_0(u_1-u_2)}\cdots\Delta(u_m)\, e^{-K_0 u_m} E_0,$$

absolutely convergent for bounded Δ (the m-th term is bounded by $(\int\|\Delta\|)^m/m!\,\|E_0\|$), so the series is the solution, not an approximation. Every term is a product of lazy propagators and increments, and every increment is built from the stage

cores of Theorem 4, so $E(T)$ lies in the span generated by the lazy flow of E_0 and the stage targets. The first-order term paired with the lazy solution is $-\int_0^T \langle E_{\text{lazy}}(2T-u), \Delta(u) E_{\text{lazy}}(u)\rangle\, du$, whose sign is fixed by two properties at each u: $\Delta(u) \succeq 0$ on the relevant subspace (the sign lemma below) and the alignment of $E_{\text{lazy}}(u)$ with $E_{\text{lazy}}(2T-u)$ in the modes Δ touches (persistence; automatic in a fixed basis). Under both, every odd term reduces the energy and the even terms are controlled by the preceding odd ones, so $\|E(T)\|^2 \le \|E_{\text{lazy}}(T)\|^2 - 2\int_0^T \langle E_{\text{lazy}}(2T-u), \Delta(u) E_{\text{lazy}}(u)\rangle du + O(\|\Delta\|^2)$, with the fixed-basis case resumming exactly to the mode-wise bound of Section 1. The expansion thus identifies the sign lemma and persistence as the complete set of hypotheses, order by order, and shows there is no third.

What the data say about the two hypotheses. On the reference run the sign condition holds on the entire support of the late residual, because every direction carrying it satisfies the feature-change inequality of Theorem 2 [n=200]; what fails is persistence in the initial basis: the subspace carrying the residual at $t = 44$ is only 55% retained at $t = 428$, the accumulated increment on those modes being 10^2–10^3 times the initial gaps. Persistence is therefore false there, not merely unproved, and the error statement must be organised in the instantaneous eigenframe, where the residual's rank profile moves slowly (the front of Section 1) even though its direction does not; the fixed-basis bounds are then the adiabatic limit and the rotation enters as a named remainder, coupling over current gap, per stage. That derivation is the remaining step.

The moving-frame identity. With $K(t) = V\Lambda V^\top$ and $c = V^\top E$ the residual in the instantaneous eigenframe, $\dot{c} = -\Lambda c - Ac$ with $A = V^\top \dot{V}$ antisymmetric, so $\frac{d}{dt}\|c\|^2 = -2c^\top \Lambda c$ exactly: rotation redistributes energy among the current modes and does not change the norm. If the sorted eigenvalues are nondecreasing from t_0 on, which follows from the stage increments being positive on the relevant subspace (discharged by Theorem 2 on the support of the late residual), then at every t the descent rate is at least the frozen-kernel rate evaluated on the *current* rank profile, with no hypothesis on eigenvector persistence. What a closed comparison with the lazy trajectory then needs is the evolution of the rank profile, i.e. the moving cutoff of Section 1: in the moving frame persistence is a statement about where the residual sits, not which way it points, and the fixed-basis bounds are the case $A = 0$.

The cutoff bound. Fix the index set $u = \{k : \lambda_k(T)T < 1\}$ of modes still unfit at T, with cutoff rank $k^* = \min u^c$, and transport it along the eigenvector flow by continuity (through crossings, which are massive here: sorted rank is not an invariant label), $P(t) = V(t)\Pi_u V(t)^\top$. With $E_u = PE$, $E_f = (I - P)E$ and the cross-front rotation rate $\rho = \max_{k\in u, j\notin u} |A_{kj}|$,

$$\tfrac{d}{dt}\|E_u\|^2 = -2E_u^\top K E_u + L, \qquad \tfrac{d}{dt}\|E_f\|^2 = -2E_f^\top K E_f - L, \qquad |L| \le 2\rho\|E_u\|\|E_f\|,$$

the leakage L being the only coupling between the two subspaces. Hence, with λ_u, λ_f the extreme eigenvalues of K on them,

$$\|\Pi_u c(t_0)\|\, e^{-\int_{t_0}^T \lambda_u} - \int_{t_0}^T \rho\|E_f\| \;\le\; \|E(T)\|, \qquad \|E(T)\|^2 \le \|\Pi_u c(t_0)\|^2 + 2\int_{t_0}^T \rho\|E_u\|\|E_f\| + \|E_f(T)\|^2,$$

with $\|E_f(T)\| \le \|(I - \Pi_u)c(t_0)\|e^{-\int \lambda_f} + \int \rho\|E_u\|$. At $t_0 = 0$, $\Pi_u c(0)$ is the energy of the initial error on the init-time eigenvectors that evolve into the unfit set at T (not the energy above sorted rank k^* at $t = 0$, which mislabels the modes by an order of magnitude here), the cutoff being set by the *lifted* spectrum at T; computing it requires the continuously tracked eigenvectors, which the chain carries. Rotation enters only through the leakage, which is weighted by the fit residual and therefore decays: rotation among unfit modes, where Corollary (iii) above places most of it, moves energy within P and costs nothing. No persistence hypothesis appears; monotonicity of the sorted spectrum (Weyl, from the discharged sign condition) is the only input. The cutoff rank is bracketed two-sidedly by Theorem 2 through the accumulated feature change, ρ is the per-stage coupling over current gap across k^*, and the static held-rough floor of Section 1 is the case in which the cutoff is read from the frozen spectrum: the lift moves the cutoff up in rank, and that is the whole effect of feature learning on the error, in one scalar. For a direction v, the time it leaves u solves $\lambda_v(T_v)T_v = 1$ and is bracketed by $(\|D_v(t)\| \mp \ell_v)^2 t/n = 1$: the delayed-feature-learning event as an exact crossing condition with a two-sided bracket from the accumulated drive and a sharpness set by ρ at the crossing.

The rate comparison, and the one lemma it rests on.

Lemma 1 (Sign of the cross term; status: derived for the core part, measured for the rest)**.** $v^\top \mathrm{Cross}_s v = \frac{2}{n}\big[\langle \Psi_0^\top v, \delta\Psi_s^\top v\rangle + \sum_{r<s}\langle \delta\Psi_r^\top v, \delta\Psi_s^\top v\rangle\big]$. *The second term's core-core part is nonnegative while the integrated residual keeps its direction (shared drive); the first term is measured negative at $t = 0$ for raw residuals and sign-free for centred ones, positive after the transient, and is not derived.*

If the lemma holds from t_0 on, then $\dot{L}(t) \leq -\langle E, K(t_0)E\rangle$ for $t \geq t_0$: feature learning never slows the descent below what the kernel at t_0 would produce on the current residual. A closed-form bound on $\|E(T)\|$ needs in addition a persistence condition on the residual direction, because $K(t) \succeq K(t_0)$ does not control mode-wise decay when the eigenvectors rotate, and they rotate by about half in the relevant band by mid-training [n=200].

As a theoretical object the chain is exact: infinitely many expansion points, each to all orders, is the flow itself, and everything above uses only that. How a truncation at one stage reaches T (the error accounting that makes every reduced version of the chain a controlled approximation), and the numerical verifications of the identities and of the leading orders, are collected in Appendix A.

Term-wise corollaries. Once the right series is used, several statements become unconditional. (i) Direct is the richness series at argument $\Delta\epsilon$, so its k-th lift term is Δ^k times Euler's: the lift comparison holds order by order inside the radius with no sign lemma. (ii) By the variational principle $\lambda_{\max}(K_0 + \Delta) \geq \lambda_1 + v_1^\top \Delta v_1$ and $\lambda_{\min}(K_0 + \Delta) \leq \lambda_{\min} + v_{\min}^\top \Delta v_{\min}$ for any symmetric Δ, so the totality of the higher-order terms pushes the top eigenvalue up and the bottom down, exactly and with no convergence question (the Rayleigh–Schrödinger series itself converges for $\|\Delta\|$ below half the eigenvalue's gap, which the chain's stages satisfy for the top and only at fine resolution for the tail); its second-order term $\sum_{k>1} |v_1^\top \Delta v_k|^2/(\lambda_1 - \lambda_k) \geq 0$ is the leading piece: progressive sharpening of the top mode has an unconditional second-order component, and the deep tail an unconditional second-order depression, which is the measured early decrease of the kernel along init-tail directions before the first-order lift overtakes it. (iii) Mode k rotates by $\sum_{j\neq k}(v_j^\top \Delta v_k)/(\lambda_k - \lambda_j)\, v_j$, coupling over gap; with an init spectrum $\sim k^{-a}$ the relative gap is $\sim a/k$, so the rotation grows with rank, i.e. with roughness: the fixed-basis assumption fails exactly where the roughness theory places the action, and |coupling|/|gap| is a computable criterion for when a direction keeps its identity. (iv) Two initialisations with the same unit statistics differ in every Dyson term only through the propagation part of the stage features, so the spread of $E(T)$ over initialisations is bounded by the propagator times the accumulated propagation mismatch: the irreducible uncertainty of any statistical description is an explicit, measurable, width-dependent quantity, and whether the statistical chain becomes exact at large width is the question whether that quantity shrinks with width.

2.6 Consequences for the held rough error

The results above replace the closure of Section 1 and sharpen its exact part. (i) Proposition 1 is the energy identity projected on a fixed P; the cutoff bound is the same identity with P transported along the eigenvector flow, the static off-diagonal leak becoming the cross-front rotation, and the fixed-P statements are its $A = 0$ reading. (ii) Proposition 15(b) is the Hellmann–Feynman lift bound; its worst-case integral $\Sigma(s) = \sigma_0 + as$ is replaced by the two-sided trajectory bracket of Theorem 2 on every eigenvalue of PKP. (iii) The efficient-supply closure of the earlier draft must be dropped: along training the Rayleigh quotient of the kernel on the rough residual is not a fixed fraction of σ_R^2 but decays, $c(t) \approx \lambda_{\text{eff}}(t)/\sigma_R(t)^2 \to 0$, because the lift raises the top of the rough subspace while the residual moves to its slowest directions, the front [n=200: $c = 0.02$–0.09 at $t = 0$, 0.001–0.005 late, with σ_R^2 lifted 6–50×]. The two-sided band built on it is replaced by the cutoff bound, whose only remainder is the leakage. (iv) The error is not held in any fixed roughness seminorm: the kernel lifts the Laplacian-rough directions and fits them, the whole residual becomes Laplacian-rough while decaying, and what is held is the energy on the modes still unfit in the *current* basis. Roughness must be measured against the lifted kernel, which is what the moving frame does. (v) The a priori cutoff of Definition 1 becomes the transported cutoff read from the lifted spectrum; Proposition 2 remains a valid a priori floor. (vi) Proposition 20 is the radius theorem at a single expansion point and the chain is its continuation. (vii) The reduced model of ignition is subsumed by the richness expansion and the crossing condition and can be cut.

2.7 Why the late stage is not predictable from the initialisation

Proposition 5 (Determinism; no chaos)**.** *For analytic N_θ the gradient-flow trajectory is real-analytic in t, so the history on any interval determines $E(T)$. The loss is a strict Lyapunov function, so there are no recurrent sets and no sustained exponential separation of nearby trajectories; what remains is finite-time stretching along transiently negative-curvature directions and algebraic amplification through the lift. [n=200: over the full run a 10^{-7} relative perturbation of θ_0 grows by a factor 6 (random direction) and 21 (tail-feature direction), while a top-feature perturbation is contracted 16× during the fit of the top mode and regrows to 5× through the lift; the induced residual perturbation grows by two to three orders of magnitude and ends*

at 10^{-6} relative. The growth is algebraic throughout, roughly $t^{0.2\text{–}0.4}$ after $t \approx 10$, with no exponential phase in 3.5 decades of time.]

So the obstruction is conditioning, and it has three parts, each a theorem.

Theorem 8 (Hidden state)**.** *Let A and Δ be the input activations and the outgoing backprop signals of a hidden-hidden block on the data at θ_0. For every $N = UV^\top$ with $U^\top\Delta = 0$ and $V^\top A = 0$ (such N exist whenever the widths exceed the ranks, e.g. $H > n\,n_y$, and have any norm), the initialisation $\theta_0 + N$ has the same unit states, output, residual, loss, gradient and kernel on the data, and differs only in the propagation part of Proposition 3(d). Hence $E(T)$ is not a function of (K_0, E_0, Y), nor of any finite number of their time derivatives at 0 when the widths allow the corresponding null spaces. [Verified: with $\|N\|_F = \|W\|_F$ the states agree to $5 \cdot 10^{-15}$ and the kernel increments after one step differ by 1%.]*

Proposition 6 (Visibility)**.** *Let B be the modes of $K(t_0)$ carrying a fraction ϕ of $\|E(T)\|^2$. Their share of the loss-decrease rate at t_0 is at most $\max_B \lambda_k(t_0)/\lambda_{\text{eff}}(t_0)$, and in the fixed-basis solution their components are constant on $[0, t_0]$ to relative order $\lambda_k(t_0)t_0$. [n=200: median $1.4 \cdot 10^{-6}$ at $t_0 = 0.01T$ and 10^{-2} at $t_0 = 0.1T$; share of the rate $8 \cdot 10^{-7}$ and $7 \cdot 10^{-4}$.]*

Proposition 7 (Amplification)**.** *The same modes are later lifted by the gain of Theorem 2 (measured 10^2–10^4), coherently across blocks, quadratically in the accumulated drive, and $E_k(T) \sim \exp(-\int \lambda_k)$ is exponentially sensitive to the lifted eigenvalue. In the fixed-basis model the sensitivity of $\log|E_k(T)|$ to a mode's growth exponent is $2\Lambda_k(T)\log(T/t_0)$ with $\Lambda_k(T) \approx 1$ for the modes that matter.*

Corollary 3 (The conditioning statement)**.** *To predict $E(T)$ to relative precision ε from the history up to t_0 the tail components must be known to relative precision of order $\varepsilon\,\lambda_k(t_0)t_0$ divided by the amplification, i.e. about 10^{-7} at $t_0 = 0.01T$ for $\varepsilon = 0.1$ [n=200]: five orders of magnitude below the initialisation-to-initialisation variation of the early history ($2 \cdot 10^{-2}$) and below single precision. The late state is a smooth, deterministic function of the early state whose relevant derivatives are enormous exactly in the directions that are unobservable early.*

The three parts have a common root: the drive is the residual, the residual lives in the slow modes of the current kernel, and the slow modes are both the ones that have not moved yet (invisible) and the ones feature learning targets (amplified). Feature learning is the mechanism by which the network

amplifies precisely the information that no early observable contains, and that information is seeded by the initialisation through Theorem 8. This is why every early predictor we tried (frozen kernel, gain-law extrapolation, residual front, the loss itself) orders designs within 1.1× at chance and within 1.3× at 65–76%, why seed pairs that agree to 2% for the first tenth of training end 15% apart, and why the only object that reproduces the late error is the trajectory carried forward [n=200].

2.8 What a statistical description can promise

The impossibility above is a statement about one network. A statistical description does not predict one network; it predicts the distribution over initialisations, and by Theorem 8 that is the correct object, because the quantity that defeats prediction is the draw of the hidden state. Three consequences follow, two of them proved and one to be measured.

The irreducible uncertainty of any statistical prediction for a single run is the spread of the late error over initialisations, 15% median here [n=200]. A design comparison finer than that spread is not well posed for one run of each design, whatever the method; it is well posed for the expected error, which is a statistic.

The coherent chain is the statistical approximation: replace the network's own units by draws from the unit distribution and the propagation part by its average, and Propositions 3(b,c) and Theorem 4(a) close on the data. On the one case run to convergence [H=96, depth 2, n=200] it reproduced the structure, the gain law and the ordering of designs, with a 2× bias in the late error and a 4–5× excess in the tail gain traced to the averaged propagation term and to the transfer of the ensemble's lift; at depth 3 the bias was larger (5× in the tail gain). It was not run at larger width. Whether the bias and the seed spread decrease with width, as the propagation term being a finite-width fluctuation suggests, or persist with depth, as the depth-3 result warns, is the single most consequential unmeasured quantity in this work, and it decides the question in the next paragraph.

The diagnostic that is well posed is the expected late-stage ability of an architecture. If the spread over initialisations shrinks with width and the coherent chain converges to the exact one in that limit, then ranking architectures by their expected late error is both meaningful and computable without the particular initialisation, and the finite-width exponent of the convergence is the error bar. Everything in this section is set up for that measurement: the exact chain for the per-initialisation truth, the coherent chain for the expectation, and the seed spread for the floor. The ranker built

from the identities (lazy error, H_2, top eigenvalue, output scale) reached a 1.3× resolution out of sample [n=200], which is the expected accuracy of a one-Jacobian statistic and should be read as a lower bound on what the expected-error ranking can do.

2.9 Scope and what to measure

Every number in this section comes from $n = 200$ training points on the Lorenz-63 one-step map, at which the training set is nearly fit within the budget, the residual-adapted late phase is short, the depth-6–8 "floor" is a sample-size artefact, and the negative lazy-transfer sign in the validation split is overfitting of a 200-point sample. The theorems carry n as a symbol and are unaffected; the verifications, bulk ratios, exponents, signs, transfer fractions and the ranker's fits are all $n = 200$ statements and are to be redone at a sample size that leaves a genuine late stage before any is quoted. The exact chain does not scale to large n (one $N \times N$ eigendecomposition per stage); the ground-truth runs do, so the redo is training-only, and it should include, at that n: the seed spread of the late error against width and depth; the gap between the own-initialisation and the other-initialisation chain against width and depth; the cross-term sign and the spike bound along the trajectory; the validation split; and the Euler-versus-direct comparison at several Δ from a shared initialisation.

Hyperparameter transfer (negative result, n=200). A sweep over widths 48–384 and output scales 0.25–2 at fixed budget $3000/s^2$ did not bracket a training-error optimum (the richest scale was best at every width and depth) and the feature-to-lazy ratio of the descent rate did not collapse across widths under either the standard or the μP-like labelling; at fixed scale it grew with width. On training error at fixed budget, richness has no cost here, so the transfer question needs a criterion with an optimum (test error or a stability constraint) and is left open.

A The chain under truncation: error propagation and verifications

A.1 How a truncation at one stage reaches time T

A truncated chain replaces the exact stage map by its order-k expansion about the stage's initial state. Three exact structures decide how the local error of stage s appears at T.

Proposition 8 (The propagator is algebraic)**.** *A perturbation of the state at time s evolves by $\dot{\delta\theta} = -H(t)\,\delta\theta$, $H = J^\top J/n + \frac{1}{n}\sum_i E_i \nabla^2 N_i$, so the propagator $\Phi(T,s) = \mathrm{Texp}(-\int_s^T H)$ obeys $\|\Phi(T,s)\| \le \exp\big(\int_s^T \lambda_{\max}(-H)_+\big) \le \exp\big(\int_s^T \frac{1}{n}\|E\|H_2\big)$: only the residual-curvature part can expand, and its integrand decays with the residual. [n=200: $\|\Phi\| \le 21$ over a full run in the worst direction, i.e. an integrated negative curvature $\ln 21 \approx 3$, an average of 10^{-3} per unit time; the H_2 form of the bound is valid but loose by e^{13}, since it replaces the negative curvature by its worst case $\|E\|H_2/n \approx 10^{-2}$.]*

Proposition 9 (Three channels)**.** *A feature error $\delta\Psi_s$ at stage s reaches T as (i) an additive, undamped feature offset $\Phi(T,s)\delta\Psi_s$ (since $\Psi_T = \Psi_0 + \sum_r \delta\Psi_r$), hence a kernel error linear in the offset while it is smaller than the features; (ii) a residual error $\delta E(T) = -\int_s^T \mathrm{Texp}(-\int_u^T K)\,\delta K(u) E(u)\,du$, contracted by K in every lifted direction and sourced only by the kernel error acting on the residual; (iii) a feedback on later stages, $\delta(\delta\Psi_r) \sim J'[J^\top \delta E]$ for $r > s$, bounded by Proposition 8 and of the order of (ii) times the drive.*

Theorem 9 (Global error of the order-k chain)**.** *If each stage's order-$(k{+}1)$ remainder is at most ε times its first-order increment, then $\|\Psi_T^{\mathrm{chain}} - \Psi_T\| \le \varepsilon\,\Phi_{\max}\sum_s \|\delta\Psi_s\|$ and $\|E_T^{\mathrm{chain}} - E_T\| \le \frac{2\varepsilon\Phi_{\max}}{n}\|\Psi_T\|\sum_s \|\delta\Psi_s\| \int_0^T \|E\| + O(\varepsilon^2)$. The stage errors are proportional to the stage* increments *and add, so their sum is ε times the total feature change along the path, not ε times the number of stages: the chain is consistent of order k and stable with an algebraic propagator, hence convergent, uniformly in the number of stages. Fewer, longer stages at higher order and many short stages at first order both converge; they differ in cost, with $S_k \sim \varepsilon^{-1/k}\int(\|\Psi^{(k+1)}\|/\|\Psi'\|)^{1/k}$ stages needed.*

Proof. Triangle inequality over stages with Proposition 9(i), Proposition 8 for the propagator, variation of constants for (ii). The numerical convergence to three digits across a tenfold range of stage budgets at $k = 1$ is this theorem observed. □

Remark 2. For proofs the $k = 1$, many-stage end suffices: it is the flow itself, and nothing in this section needs more than first order per stage plus the propagator bound. For computation the chain is not cheap: long stages need high-order directional derivatives of the Jacobian at every expansion point, short stages need of order 10^4 of them; either way its cost is that of integrating the dynamics. Its value is the error accounting, which makes every truncated or reduced version of it a controlled approximation with a named error term. The constants are finite but not small in exactly two circumstances: early, where the residual-curvature part of H makes the

propagator grow (the sharpness-reduction phase), and for errors landing in a slow mode, where (ii) gives no damping and the offset of (i) is later amplified by the lift of that mode. That is the chain's own statement of Propositions 6–7: what is wrong, or unknown, in the slow modes early survives and is amplified; what is wrong in the fast modes is forgotten.

A.2 Numerical verifications of the identities and of the leading orders

Remark 3. A numerical realisation of the chain (one eigendecomposition of the $N \times N$ kernel per stage, the maps of Proposition 3(c,d) applied to the network's own units) reproduces the real gradient-flow run to four digits in $\|E\|$ and to 0.1% in every kernel eigenvalue at $t = 8$, tracks it to 1–3% at T on every design tested including depth 8 and width 512, and is converged to three digits across a tenfold range of stage budgets [n=200]. It is gradient flow on the sample integrated in large stages; its only use in this section is as the exact reference against which every reduced statement is checked.

Remark 4 (What was measured, n=200). (a) is verified to within 10% at initialisation for tanh, ReLU, GELU and softplus; the bulk is 2–5% of the core for tanh at unit gain and 30–50% for ReLU (whose local derivative is a 0/1 pattern), yet 98–100% of the first-order change lies in three singular directions for every activation. Along the path the independence of (a) fails within a few steps, as it must once the units adapt to the residual, while the rank persists: at $t = 0.3$ the per-unit-mean split gives a bulk ratio of 0.9 against the dispersion prediction of 0.05, and the cumulative change is still 98% in three directions; the correlation-free bound (b) holds with $r = 1$–2, i.e. the update lies 93% in 3 and 99.9% in 9 directions. Cores of successive stages add coherently while the drive keeps its direction, which is the cross-layer alignment measured earlier (randomising one layer's rotation destroys the lift); the bulk-plus-spike spectrum of trained weights is the random initialisation plus this coherent low-rank sum.

Remark 5 (Verification, n=200). With $\epsilon = 1/s$ and the coefficients through second order (including the second-order weight change and the feedback of the lifted kernel on the lazy residual), the predicted lift of init eigendirections agrees with the measured one to 0–16% at output scales $s = 2, 4, 8, 16$ for as long as the measured lift is below order one, including the early *decrease* along deep-tail directions and its sign change; the truncated series fails sharply once the lift exceeds one, where Theorem 6 says it must. The leading coefficient a_1 changes sign along τ, so no universal power of s

describes the delay to a given lift; the function is predicted, the exponent does not exist.